

Data Mining

- Extracting useful, hidden, and previously unknown patterns from large datasets.
- Uses automatic or semi-automatic analysis techniques.
- Helps discover meaningful information for decision-making.

Prediction Methods

- Use known variables to predict unknown or future values.
- Focus on forecasting outcomes.

Description Methods

- Find patterns that describe and summarize data.
- Results are usually easy for humans to understand.

Clustering

- Groups similar objects together.
- Objects in different groups are dissimilar.

Applications:

- Customer profiling for marketing.
- Grouping similar documents.
- Grouping genes/proteins with similar functions.
- Data summarization and size reduction.

Association Rules

- Discover relationships between items in data.
- Generate rules that predict item occurrence based on other items.

Applications:

- Market basket analysis.
- Sales promotion and shelf management.
- Linking symptoms to diseases.

Classification (Predictive Modeling)

- Builds a model to assign records to predefined classes.
- Predicts categorical outcomes.

Applications:

- Fraud detection.
- Land cover classification.
- News categorization.

- Tumor classification (benign/malignant).

Anomaly Detection

- Identifies unusual or abnormal data points.
- Also called Outlier Detection.
- Finds observations that differ significantly from normal behavior.

Regression

- Predicts a continuous numeric value.
- Uses relationships between variables.

Applications:

- Predicting sales from advertising spending.
- Predicting wind speed from weather factors.

Motivating Challenges

Scalability

- Handle very large datasets efficiently.
- Maintain performance with billions of records.

High Dimensionality

- Data may contain hundreds or thousands of attributes.
- Increases computational complexity.

Heterogeneous & Complex Data

- Data can be text, images, graphs, or time-series.
- Requires specialized analysis methods.

Data Ownership & Distribution

- Data is often stored in different locations or organizations.
- Raises privacy and accessibility issues.

Non-traditional Analysis

- Traditional statistical methods may be insufficient.
- Requires advanced data mining techniques.